

Gender and Socioeconomic Inequality in Science Achievement

A Bayesian Multilevel Analysis of TIMSS 2023 England Grade 8

ST308: Bayesian Inference

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1 Introduction

Academic performance in secondary education shapes future education and employability. This project investigates two key dimensions of educational inequality: gender and socioeconomic disadvantage, and whether their effects on science achievement vary across schools with different levels of economic disadvantage. Using (TIMSS , 2023) data for England, a Bayesian multilevel framework is adopted to capture both within-school and between-school variation. The research question is: *How do gender and socioeconomic background influence science achievement among Grade 8 students in England, and do these effects vary across schools with different levels of economic disadvantage?*

2 Data and Exploratory Analysis

2.1 Dataset Overview

The following data from TIMSS 2023 is obtained:

Table 1: Variable descriptions

Variable	Description	Levels
PV1SCIE_std	Standardised science score (response variable)	Continuous
GENDER	Student gender	0 = Girl, 1 = Boy
ESCS	Home educational resources	1 = Few, 2 = Some, 3 = Many
BOOKS	Books at home	1 = 0–10, 2 = 11–25, 3 = 26–100, 4 = 101–200, 5 = 200+
INTERNET	Internet access or own room	1 = None, 2 = Either, 3 = Both
BELONGING	Sense of school belonging	1 = Little, 2 = Medium, 3 = High
SCH_DISADV	School % of economically disadvantaged students	1 = 0–10%, 2 = 11–25%, 3 = 26–50%, 4 = 50%+
ACAD	School academic emphasis	1 = Medium, 2 = High, 3 = Very High
SCH_SES	School SES composition	1 = More Affluent, 2 = Mixed, 3 = More Disadvantaged
PAR_EDU	Parental education level	1 = University+, 2 = Post-secondary, 3 = Upper secondary, 4 = Lower secondary, 5 = Some primary

After removing 1,184 students from 28 schools that did not return the school questionnaire, the final dataset contains 3,055 students across 99 schools.

The response is standardised science score (PV1SCIE_std). Student-level predictors are gender, home educational resources, books at home, and school belonging. School-level predictors are percentage of economically disadvantaged students, academic emphasis and school SES.

PAR_EDU was excluded due to 39.3% missingness. For variables with under 5% missingness, missing indicator flags were created and values replaced with zero, preserving the full sample of 3,055 students across all models.

Table 2: Missingness summary for key variables

Variable	<i>n</i> missing	%
PAR_EDU	1,201	39.3
BELONGING	121	4.0
Internet_or_room	94	3.1
ESCS	81	2.7
BOOKS_HOME	72	2.4

2.2 Exploratory Data Analysis

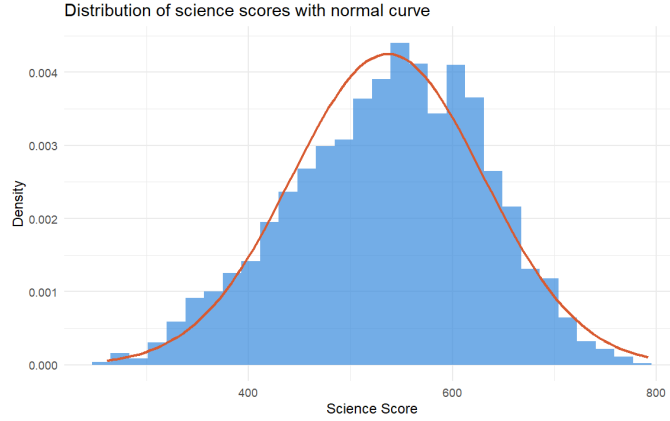


Figure 1: Distribution of science scores with normal curve overlay.

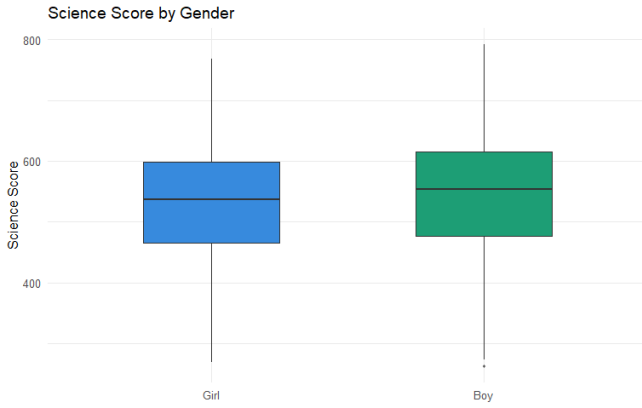


Figure 2: Science scores by gender.

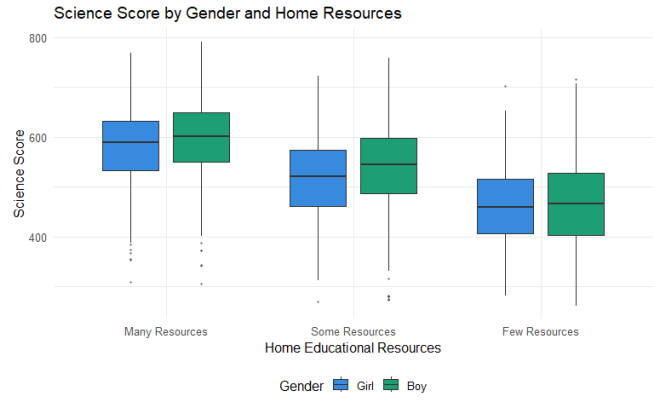


Figure 3: Science scores by gender and home resources.

Figure 1 shows science scores are approximately bell-shaped with a mean of 536 points, with the normal curve overlay confirming the distribution is approximately Gaussian. This supports the use of a Gaussian response model. Figure 2 shows boys outscore girls by approximately 16 points (median: 537 vs 553), though there is substantial overlap in the distributions. Figure 3 reveals that the gender gap seems to be consistent across all levels of home educational resources. Students with many home resources score approximately 150 points above those with few resources.

3 Model Fitting

A staged modelling approach was adopted: frequentist linear regression as a baseline, followed by Bayesian linear regression, and then Bayesian multilevel models. This allows direct comparison of explanatory power, uncertainty quantification, and treatment of the hierarchical school structure.

3.1 Frequentist Linear Regression

As a baseline, a full linear regression model was fitted with all candidate predictors and missing indicator flags:

$$PV1SCIE_std_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

where $\mathbf{x}_i$ is the vector of explanatory variables for student i . Backward elimination (at 5% significance) removed `INTERNET`, `SCH.SES_disadv`, `INTERNET_missing`, and `ESCS_missing`. The final model retains gender, home resources, books, belonging, school disadvantage rate, academic emphasis, mixed school type, and missing indicator flags for belonging and books.

Table 3: Frequentist linear regression: final model coefficients

Predictor	Estimate	Std. Error	<i>t</i> value	<i>p</i> value
(Intercept)	-1.296	0.106	-12.20	<0.001
GENDER	0.167	0.031	5.42	<0.001
ESCS	0.296	0.034	8.77	<0.001
BOOKS	0.174	0.020	8.85	<0.001
BELONGING	0.131	0.024	5.50	<0.001
SCH_DISADV	-0.153	0.020	-7.58	<0.001
ACAD	0.113	0.026	4.28	<0.001
SCH_SES_mixed	0.064	0.033	1.98	0.048
BELONGING_missing	-0.248	0.104	-2.38	0.018
BOOKS_missing	1.030	0.128	8.07	<0.001
Adjusted $R^2 = 0.310$. Residual SE = 0.831 on 3045 df.				

The final model explains 31% of the variation in science scores (adjusted $R^2 = 0.310$). Measures of socioeconomic background are the dominant predictors: each unit increase in home educational resources (ESCS) adds 0.30 standard deviations, and each additional level of books at home (BOOKS) adds 0.17 standard deviations. The gender gap is smaller at 0.17 standard deviations, while school disadvantage reduces scores by 0.15 standard deviations per level. Sense of school belonging and academic emphasis contribute more modestly (0.13 and 0.11 standard deviations respectively). The large positive coefficient on BOOKS_missing (1.03) is likely a data artefact and is treated as a control rather than a substantive finding. The model assumes student independence regardless of school, motivating the multilevel approach that follows.

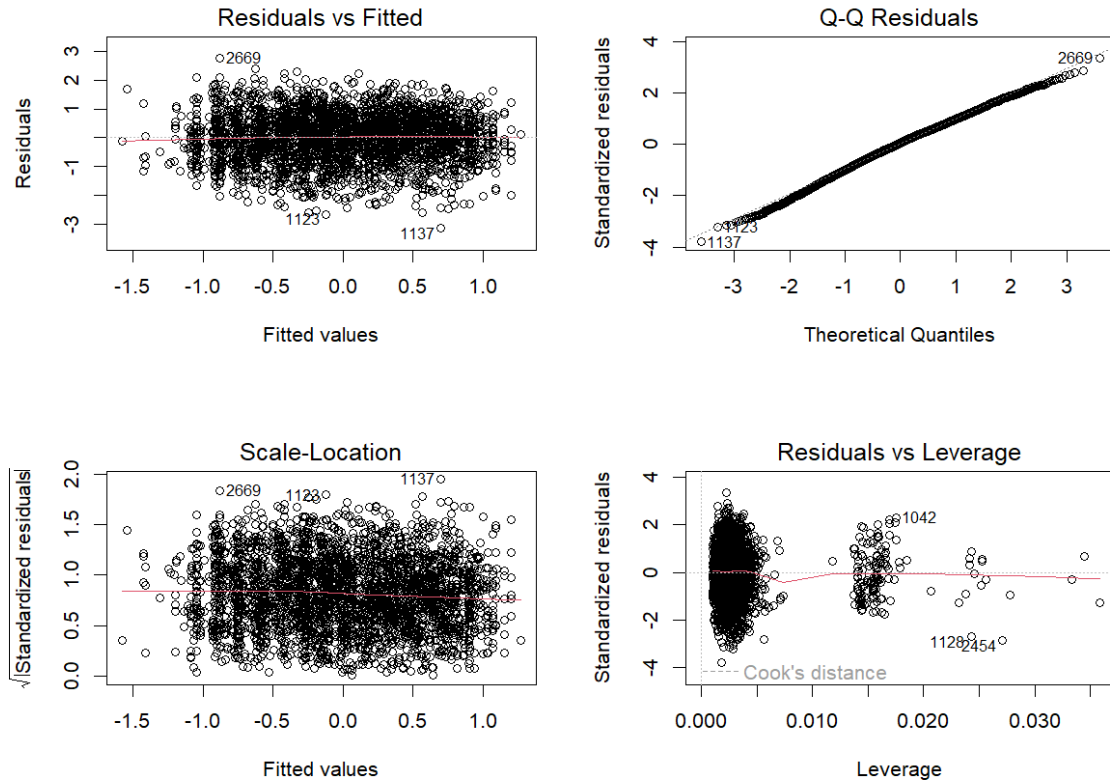


Figure 4: Frequentist model diagnostic plots.

Figure 4 shows the model assumptions are reasonably satisfied. The residuals versus fitted plot shows no clear structure, supporting linearity. The Q-Q plot shows minor tail deviations but the central distribution remains approximately normal. The scale-location plot indicates mild heteroscedasticity, not at a level that invalidates inference. The residuals against leverage plot shows no highly influential observations.

The main limitation of the frequentist model is that it assumes all students are independent, which is unlikely given school clustering. This motivates the Bayesian and multilevel formulations that follow.

The frequentist linear regression is modelled as the following:

$$\begin{aligned} \text{PV1SCIE_std}_i = & -1.30 + 0.17 \text{ GENDER} + 0.30 \text{ ESCS} + 0.17 \text{ BOOKS} + 0.13 \text{ BELONGING} \\ & - 0.15 \text{ SCH_DISADV} + 0.11 \text{ ACAD} + 0.06 \text{ SCH_SES_mixed} \\ & - 0.25 \text{ BELONGING_missing} + 1.03 \text{ BOOKS_missing} + \varepsilon_i \end{aligned}$$

Non-Technical Summary: Boys outscore girls in science, but the gap is small. A much bigger factor is how wealthy a student’s home background is. Students with more books and better home resources score substantially higher than those without. Schools serving more deprived communities also tend to have lower scores overall. This suggests that poverty, is the main driver of inequality in science achievement.

3.2 Bayesian Linear Regression

The frequentist model was re-estimated in a Bayesian framework, treating parameters as random variables:

$$\begin{aligned} \text{PV1SCIE_std}_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \\ \beta_k \sim \mathcal{N}(0, 2.5), \quad \beta_0 \sim \mathcal{N}(0, 5) \end{aligned}$$

The frequentist model was re-estimated in a Bayesian framework using weakly informative Normal priors, centred at zero but allowing moderate effects to emerge from the data. Science scores were standardised prior to modelling so that the prior scales are appropriate. All nine predictors had 95% credible intervals excluding zero, so no variables were dropped.

Table 4: Bayesian linear regression: posterior means and 95% credible intervals

Predictor	Mean	2.5%	97.5%
GENDER	0.17	0.11	0.23
ESCS	0.30	0.23	0.36
BOOKS	0.17	0.14	0.21
BELONGING	0.13	0.08	0.18
SCH_DISADV	−0.15	−0.19	−0.11
ACAD	0.11	0.06	0.16
SCH_SES_mixed	0.06	0.00	0.13
BELONGING_missing	−0.25	−0.45	−0.05
BOOKS_missing	1.03	0.78	1.27
Bayesian R^2 (median) = 0.312. All $\hat{R} = 1.0$.			

Bayesian $R^2 = 0.312$, closely matching the frequentist adjusted R^2 of 0.310. The main benefit of the Bayesian framework is improved uncertainty quantification: credible intervals directly express the probability that each effect lies within a given range, unlike frequentist confidence intervals.

Non-Technical Summary: The Bayesian model produces the same findings as the frequentist model. However, both models so far have assumed that students are completely independent of one another, which is unlikely since students within the same school share teachers, resources, and environment. The next model accounts for this by allowing each school type to have its own baseline score.

3.3 Bayesian Multilevel Model: Random Intercepts

SCH_DISADV was selected as the grouping variable on both theoretical and empirical grounds. Theoretically, it most directly captures the school-level poverty dimension central to the research question. Empirically, ICC calculations from null models showed SCH_DISADV explained the largest share of between-school variance (ICC = 0.204), compared to SCH_SES (ICC = 0.106) and ACAD_EMPHASIS (ICC = 0.139).

A Bayesian multilevel model was therefore fitted with students nested within school disadvantage bands, allowing each band to have its own baseline science score:

$$\text{PV1SCIE_std}_{ij} = \beta_0 + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + u_j + \varepsilon_{ij}, \quad u_j \sim \mathcal{N}(0, \sigma_{\text{school}}^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

where u_j captures between-group variation in baseline performance for disadvantage band j . The same weakly informative priors were applied as before.

Table 5: Random intercept model: group-level baseline adjustments and 95% credible intervals

Group	Median	2.5%	97.5%
0–10%	0.00	−0.17	0.21
11–25%	0.00	−0.16	0.11
26–50%	0.00	−0.21	0.19
50%+	0.00	−0.21	0.45
Bayesian R^2 (median) = 0.312. All $\hat{R} = 1.0$ (fixed effects).			

All four credible intervals span zero, indicating that once the fixed effects are accounted for, the four school disadvantage bands do not have meaningfully different baseline science scores. The `SCH_DISADV` fixed effect already captures the linear trend in disadvantage, leaving little residual variation for the random intercepts to explain. Bayesian $R^2 = 0.312$, identical to the single-level Bayesian model, confirming that the grouping structure adds negligible explanatory power beyond the fixed effects.

Non-Technical Summary: When we account for students’ individual home backgrounds, the type of school they attend adds little extra information about their science score. For example, a student from a low-income household in a mostly affluent school is predicted to score similarly to a student from the same low-income background in a mostly deprived school. However, this doesn’t mean school deprivation is irrelevant. This leads us to ask a more targeted question; does the gap between boys and girls in science get larger or smaller in these schools?

3.4 Bayesian Multilevel Model: Random Slopes

Interaction tests were conducted between all student-level predictors and school-level variables to identify which random slopes were warranted. `SCH_DISADV` showed significant interactions with all four student-level predictors: `BOOKS`, `ESCS`, `GENDER`, and `BELONGING`. To obtain a parsimonious model, and given that the primary research question concerns the gender gap, only `GENDER` was included as a random slope.

The random slopes model allows both the baseline science score and the gender gap to vary across school disadvantage bands:

$$\text{PV1SCIE_std}_{ij} = \beta_0 + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + u_{0j} + u_{1j} \text{GENDER}_{ij} + \varepsilon_{ij}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{school}}), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

where u_{0j} is the group-specific intercept deviation and u_{1j} is the group-specific adjustment to the gender slope for disadvantage band j . The same weakly informative priors were applied as before.

Table 6: Random slopes model: gender slope adjustments by school disadvantage band

Group	Intercept adj	CI	Gender adj	CI
0–10%	0.00	[−0.13, 0.13]	0.00	[−0.13, 0.13]
11–25%	0.00	[−0.08, 0.10]	0.00	[−0.19, 0.06]
26–50%	0.00	[−0.11, 0.07]	0.00	[−0.17, 0.07]
50%+	0.00	[−0.12, 0.13]	0.10	[−0.03, 0.26]
Bayesian R^2 (median) = 0.313. All $\hat{R} = 1.0$.				

Fixed effects remain consistent with previous models: socioeconomic background is the strongest predictor, followed by books at home and gender. All 95% credible intervals for the gender slope adjustments span zero, indicating that the gender gap does not vary significantly across disadvantage bands. However, the direction of the adjustments is informative. In moderately disadvantaged schools (11–25% and 26–50%), the gender adjustment is slightly negative, suggesting boys’ advantage over girls is marginally smaller than average in these schools. In the most severely disadvantaged schools (50%+), the adjustment is positive (0.10) with CI: −0.03 to 0.26, suggesting boys outscore girls by more in schools where the majority of students are from economically disadvantaged backgrounds.

This directional pattern is consistent with existing literature. Research from UCL (2025) report that boys in England significantly outperformed girls in Year 9 science in 2023. One plausible mechanism is the role of gender stereotypes: in highly disadvantaged communities, traditional attitudes towards girls’ participation in STEM may be more entrenched. Dasgupta Stout (2014) show that in childhood and adolescence, masculine stereotypes about STEM, peer norms, and parental expectations combine to push girls away from science, effects that are likely amplified in lower-resource environments where female STEM role models are scarce. Additionally, girls report substantially lower school belonging than boys (42% vs 29% reporting little belonging). Since belonging is a significant independent predictor of science scores ($\hat{\beta} = 0.13$), this raises the possibility that girls’ lower school engagement partially mediates the gender gap.

Non-Technical Summary: The model tested whether the gap between boys and girls in science scores is larger or smaller depending on how disadvantaged a school is. The results show the gender gap is broadly consistent across school types, but boys appear to outscore girls by slightly more in the most disadvantaged schools. This tentatively suggests that severe school-level poverty may amplify existing gender inequalities in science, possibly because highly disadvantaged environments reinforce traditional gender norms that discourage girls from engaging with science. However, the uncertainty in these estimates is too large to draw firm conclusions.

4 Model Comparison and Prediction

All models were evaluated on a held-out test set (20% of students) after refitting on the training data (80%), with all six models evaluated on exactly the same observations to ensure comparability.

Table 7: Predictive performance across all models

Model	RMSE
Frequentist linear	0.8341
Bayesian linear	0.8341
Lasso	0.8340
Multilevel intercepts	0.8337
Multilevel slopes	0.8335
Ridge	0.8333

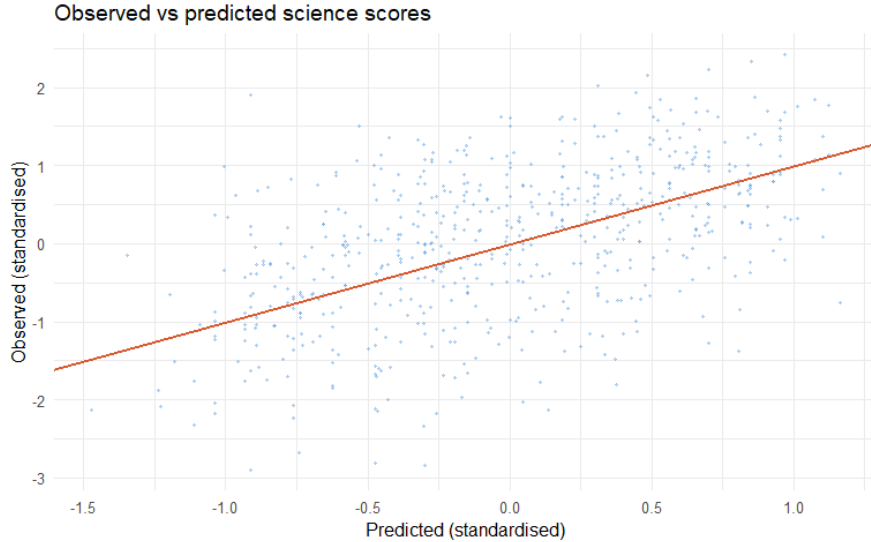


Figure 5: Observed vs predicted standardised science scores (multilevel slopes model).

All six models achieve near-identical RMSE on the held-out test set, with a total range of less than 0.001 standardised score units. Ridge performs marginally best (0.8333), suggesting mild regularisation provides a small predictive benefit. The multilevel slopes model performs second (0.8335), confirming that allowing the gender gap to vary across school disadvantage bands adds modest predictive value over the intercept only multilevel model. Lasso retained all predictors with no coefficients shrunk to zero, confirming each variable contributes meaningfully to prediction. The near identical performance across all models indicates the signal is predominantly linear and well captured by the fixed effects. Figure 5 shows the model captures the main linear trend well, though substantial residual scatter remains, consistent with the R^2 of approximately 0.31. The Bayesian multilevel slopes model is nevertheless preferred as the final specification, as it correctly represents the hierarchical data structure, quantifies uncertainty through credible intervals, and directly addresses the research question by allowing the gender gap to vary across school disadvantage levels.

5 Conclusion

This project investigated the determinants of science achievement among Grade 8 students in England using TIMSS 2023 data, with particular attention to gender and socioeconomic inequality. A staged modelling approach was adopted, moving from frequentist linear regression through Bayesian linear regression to Bayesian multilevel models with random intercepts and random slopes, allowing direct comparison of explanatory power, uncertainty quantification, and treatment of the hierarchical school structure.

Across all models, socioeconomic background emerged as the dominant predictor of science achievement. Each unit increase in home educational resources adds approximately 0.30 standard deviations to science scores, and each additional level of books at home adds 0.17 standard deviations. By comparison, the gender gap stands at 0.17 standard deviations, roughly half the magnitude of the home resources effect. School-level deprivation also independently reduces scores by 0.15 standard deviations per level of disadvantage, and academic emphasis contributes a modest positive effect of 0.11 standard deviations. This confirms that socioeconomic inequality is the primary driver of educational inequality in science achievement in this context, with gender playing a secondary but nonetheless statistically meaningful role.

The gender gap is small but consistent across all model specifications, with boys outscoring girls by approximately 0.17 standard deviations after controlling for socioeconomic background and school-level factors. Frequentist interaction tests suggested a borderline significant interaction between gender and school disadvantage. However, in the Bayesian multilevel random slopes model, all gender slope adjustments are centred at zero with credible intervals spanning zero across three of the four disadvantage bands, indicating that boys' advantage in science is broadly uniform across the educational system. The most disadvantaged schools (50%+) show a tentatively larger gender gap

(adjustment of 0.10 SD, CI: -0.03 to 0.26), which is directionally consistent with the hypothesis that severe school-level deprivation may amplify gender inequalities in science. However, the wide credible intervals mean this finding cannot be distinguished from chance and should be interpreted cautiously. The consistency of the gender gap across school types suggests that interventions targeting the gender gap in science cannot rely solely on reducing school-level deprivation, and broader structural mechanisms warrant attention.

Sense of school belonging emerged as a significant independent predictor of science scores ($\hat{\beta} = 0.13$), raising the possibility that differences in school engagement between boys and girls partially mediate the observed gender gap. This warrants further investigation through formal mediation analysis, as belonging may represent an actionable lever for reducing gender disparities in science achievement.

The multilevel models revealed that once individual-level poverty measures and school disadvantage rates are included as fixed effects, school disadvantage bands add little additional explanatory power beyond what is already captured by those fixed effects. The random intercept model produced a Bayesian R^2 of 0.312, identical to the single-level Bayesian model, confirming this. The random slopes model improved marginally to 0.313, suggesting that allowing the gender gap to vary across disadvantage bands adds a small but non-trivial amount of explanatory power. All six models achieved near-identical RMSE of approximately 0.834 on the held-out test set, with Ridge performing marginally best (0.8333) and the multilevel slopes model second (0.8335). Lasso retained all predictors with no coefficients shrunk to zero, confirming that each variable contributes meaningfully to prediction. The near-identical predictive performance across all models indicates that the signal in the data is predominantly linear and well captured by the fixed effects. The Bayesian multilevel slopes model is nevertheless preferred as the final specification, as it correctly represents the hierarchical data structure, quantifies uncertainty through credible intervals, and directly addresses the research question by allowing the gender gap to vary across school disadvantage levels.

Limitations. The analysis is cross-sectional and cannot establish causality. With only four school disadvantage bands, the multilevel structure is limited and random effects are estimated with substantial uncertainty. The variance component Σ_{school} showed $\hat{R} = 1.1$ in the random intercepts and random slopes models, indicating convergence difficulties. TIMSS does not collect direct household income data, so socioeconomic effects are measured through proxy variables which may underestimate true SES inequality. Future work could use the TIMSS longitudinal database to examine how these gaps evolve over time, and formal mediation analysis to test whether school belonging mediates the gender gap in science achievement.

References

- IEA. (2023). *Trends in International Mathematics and Science Study (TIMSS) 2023 International Database*. Boston College, TIMSS & PIRLS International Study Center. <https://timss2023.org/data>
- Dasgupta, N., & Stout, J. G. (2014). Girls and women in science, technology, engineering, and mathematics. *Policy Insights from the Behavioral and Brain Sciences*, 1(1), 21–29. <https://doi.org/10.1177/2372732214549471>
- UCL. (2025, March 10). *Boys in England significantly outperforming girls in maths and science*. UCL News. <https://www.ucl.ac.uk/news/2025/mar/boys-england-significantly-outperforming-girls-maths-and-science>